A Sharp Local-Question Threshold for GHZ-Equatorial Strategies in Four-Player XOR Games

Ziao Tang[1], Chengkai Zhu[2], Ge Bai[1], Xin Wang[1,*], and Ranyiliu Chen[3,†]

[1]The Hong Kong University of Science and Technology (Guangzhou), Guangdong 511453, China
[2]QudeLeap Research, Shanghai 200030, China
[3]Quantum Science Center of Guangdong-Hong Kong-Macao Greater Bay Area, Shenzhen 518045, China

Abstract

We determine the smallest number of active questions per player at which a four-player binary exclusive-or (XOR) game of commuting-operator value one need not admit a Greenberger–Horne–Zeilinger (GHZ) equatorial realization. Such a realization uses the four-qubit GHZ state and equatorial qubit observables, reducing perfect play to additive phase equations. We prove that every four-player XOR game with commuting-operator value one and at most three active questions per player has a perfect GHZ-equatorial strategy. Conversely, we construct a uniform eight-clause game with four active questions per player whose commuting-operator value is one but whose phase equations are inconsistent. Thus four is the sharp local-question threshold. The positive result follows by lifting every integral incidence obstruction to an ordered noncommutative refutation, using primitive circuits, forest matchings, and ternary Hamming geometry. For the separating game, a Klein four-group incidence relation obstructs the phase system, while an even-subgroup normal form and degree-one and degree-two Magnus coefficients exclude refutations of arbitrary length.

*felixxinwang@hkust-gz.edu.cn
†chenranyiliu@quantumsc.cn

Contents

1 Introduction

Nonlocal games recast Bell experiments as cooperative tests and provide a common language for Bell inequalities, entangled strategies, and operator-algebraic models of multipartite measurements [Bel64, CHSH69, CHTW04]. In a binary n-player XOR game, the verifier samples a question tuple (or a *clause*), receives one bit from each player, and checks a prescribed parity. Despite this restriction, multiplayer XOR games have served as a tractable testbed for multipartite Bell violations and quantitative bounds on entangled values [PGWP+08, BV13]. At value one, their constraints also admit exact algebraic formulations.

At value one, every positive-weight clause imposes an exact relation among the players' binary observables. We compare two realizations of these relations. A finite-dimensional tensor-product strategy assigns a Hilbert space factor to each player, whereas a commuting-operator strategy represents all observables on one, possibly infinite-dimensional, Hilbert space and requires operators belonging to distinct players to commute. The latter model contains the former and is central to semidefinite and game-algebraic approaches to nonlocal correlations [NPA08, CLS17]. Since the two correlation models differ in general [Slo19, JNV+21], it is a substantive restriction to ask when commuting-operator satisfiability is already witnessed by a specific finite-dimensional construction.

For XOR games, the relevant construction is suggested by the GHZ–Mermin paradox and the structure of multipartite full-correlation Bell inequalities [GHSZ90, Mer90, WW01]. We use GHZ-equatorial, or maximally entangled relative-phase (MERP), strategies in the terminology of

[WHKN19]. With the shared state and local measurements given by

$$|\mathrm{GHZ}_n\rangle = \frac{|0\rangle^{\otimes n} + |1\rangle^{\otimes n}}{\sqrt{2}}, \qquad M(\theta) = \cos\theta\, X + \sin\theta\, Y,$$

one has

$$\langle \mathrm{GHZ}_n | \bigotimes_{\alpha=1}^{n} M(\theta_\alpha) | \mathrm{GHZ}_n \rangle = \cos\left(\sum_{\alpha=1}^{n} \theta_\alpha \right). \tag{1.1}$$

Thus a clause c with target parity b_c is satisfied with certainty if and only if

$$\sum_{\alpha=1}^{n} \theta_{\alpha, q_c^{(\alpha)}} = \pi b_c \pmod{2\pi}. \tag{1.2}$$

When the angles are restricted to multiples of $\pi/2$, this recovers the usual X/Y relations of the GHZ–Mermin construction. Arbitrary angles extend them to continuous $U(1)$ phase data. The phase description is abelian, although the local observables need not commute; indeed, $M(\theta)M(\phi) = e^{i(\phi-\theta)Z}$. MERP nevertheless fixes the local dimension, shared state, and measurement geometry, while a general commuting-operator strategy imposes none of these restrictions. The question is whether Eq. (1.2) is complete for perfect commuting-operator satisfiability.

Prior work identifies two complementary frontiers of MERP completeness. Werner and Wolf's analysis of multipartite full-correlation inequalities with two dichotomic observables per site underlies the two-question regime: for any number of players, commuting-operator value one is witnessed by a perfect GHZ-equatorial strategy when each player has at most two active questions [WW01, WH23]. Bene Watts et al. formulated the MERP–PREF duality, which turns consistency of the phase system into the absence of an integer incidence obstruction [WHKN19]. The same work showed that completeness does not hold in general: it identified a six-player 123 game which has a perfect finite-dimensional strategy but no perfect MERP strategy. In the other direction, Bene Watts and Helton proved that every three-player XOR game of commuting-operator value one admits a perfect three-qubit MERP strategy, for arbitrary question alphabets, and obtained a polynomial-time decision procedure for perfect 3XOR [WH23]. Taken together, these results leave unresolved the smallest active local alphabet at which perfect commuting-operator satisfiability can escape the GHZ-phase model in the four-player setting. This brings us to the central question of the paper:

> *What is the smallest value of k for which a four-player k-question XOR game can have commuting-operator value one while admitting no perfect MERP strategy?*

The following theorem gives the exact answer.

Theorem 1.1. (i) *Let G be any finite four-player XOR game. If G has at most three active questions per player, then*

$$\omega_{\mathrm{co}}(G) = 1 \quad \Longleftrightarrow \quad G \text{ admits a perfect MERP strategy.} \tag{1.3}$$

(ii) *There exists a four-player XOR game G, with four active questions per player, such that $\omega_{\mathrm{co}}(G) = 1$ but no perfect MERP strategy exists. Consequently, four is the smallest number of active questions per player for which such a separation can occur.*

We remark that part (i) does not assert uniqueness or rigidity of the GHZ realization. Whether this four-question game admits a perfect finite-dimensional non-MERP strategy remains open as well.

The four-player problem is the natural next boundary case after the three-player collapse theorem of Bene Watts and Helton [WH23], but Theorem 1.1 is not obtained by a direct extension of their proof. That argument is intrinsically organized around three page projections: projector and right-inverse constructions clear two page components, while specialized gadget maps control the remaining one. With a fourth page, the same operations leave coupled residual components on several pages, and the clearing identities no longer close. Moreover, part (ii) shows that the unrestricted four-player analogue is false. A different support-level lifting mechanism is therefore needed.

The algebraic content of this lifting mechanism is most transparent in terms of two obstruction spaces. An integer dependence among the clause-incidence rows whose target-weighted parity is odd obstructs the MERP phase equations; following [WHKN19], we call it a parity-permuted refutation specification (PREF). A true refutation is stronger: it orders clause occurrences so that every player's question word freely reduces to the identity and the total target parity is odd. The known dualities say that a PREF exists exactly when no perfect MERP strategy exists, whereas a true refutation exists exactly when the commuting-operator value is below one [WHKN19, WH23, WHK23].

Every true refutation yields a PREF after forgetting the clause order. Our three-question theorem proves the converse for every four-partite ternary support: each abelian obstruction can be lifted to an ordered noncommutative refutation. The Klein four-group game in part (ii) exhibits the first possible gap: it has a PREF but no true refutation of any length.

The positive result is proved through an exact circuit-lifting theorem. The integer relation lattice is generated by primitive support-minimal circuits, and a rank bound confines each such circuit to at most ten clauses. For every primitive circuit we construct a balanced word with the prescribed absolute multiplicities, using signed-occurrence forest matchings to produce the order and ternary Hamming geometry to exclude the exceptional configurations. The ten-point extremal case is resolved by a ternary perfect-code argument. This is a uniform lifting result rather than a bounded search. For sharpness, the Klein translation pattern supplies an immediate PREF. An even-subgroup normal form, together with the degree-one and degree-two Magnus coefficients, rules out true refutations with arbitrary order and multiplicity by an explicit integral parity certificate [Mag37, MKS76].

Section 2 develops the two obstruction criteria and the Klein witness. Section 3 proves exact circuit lifting for four ternary pages. Section 4 establishes sharpness through the Cayley-game analysis and the all-length Magnus obstruction. Section 5 discusses scope and open problems.

2 Preliminaries and obstruction criteria

2.1 Games, strategies, and active supports

A finite four-player binary XOR game G consists of finite question sets $Q_1, \dots, Q_4$ and a finite family of clauses. Clause c is specified by a question tuple

$$q_c = (q_c^{(1)}, \dots, q_c^{(4)}) \in Q_1 \times \cdots \times Q_4,$$

a probability p_c, and a target parity $b_c \in \mathbb{F}_2$. On clause c, the players return bits $a_1, \ldots, a_4$ and win precisely when

$$a_1 \oplus \cdots \oplus a_4 = b_c.$$

A strategy is perfect if it wins every clause of positive probability. The commuting-operator value $\omega_{\mathrm{co}}(G)$ is the supremum of its success probability over commuting-operator strategies.

Only the positive-weight support is relevant to perfection. The resulting reduction to a uniform distribution on the supported clauses is standard in the perfect-value setting; see, for example, [WH23, Sec. 2.1.1]. We delete zero-weight clauses and merge repeated question tuples with the same target. Repeated tuples with opposite targets preclude a perfect strategy. The resulting set

$$E \subseteq Q_1 \times \cdots \times Q_4$$

is the *reduced support*. Proposition 2.1 shows that replacing its positive distribution by the uniform distribution preserves both commuting-operator value one and perfect MERP satisfiability. We may therefore work with distinct, uniformly weighted clauses whenever only these exact properties are at issue. We state the reduction with an explicit comparison inequality and include the proof for completeness.

Proposition 2.1 (Standard support and weight reduction)**.** *Let a reduced support have m clauses, and let $p_{\min}$ and $p_{\max}$ be the minimum and maximum of a strictly positive clause distribution p. For every strategy,*

$$mp_{\min}(1 - V_{\mathrm{unif}}) \leq 1 - V_p \leq mp_{\max}(1 - V_{\mathrm{unif}}), \tag{2.1}$$

where V_p and V_{unif} are its success probabilities under p and the uniform distribution. Hence $\omega_{\mathrm{co},p} = 1$ if and only if $\omega_{\mathrm{co,unif}} = 1$. Perfect MERP satisfiability likewise depends only on the reduced support and its targets.

Proof. If $\ell_c \in [0, 1]$ is the loss probability on clause c, then $1 - V_p = \sum_c p_c \ell_c$ and $1 - V_{\mathrm{unif}} = m^{-1} \sum_c \ell_c$. Bounding every p_c between $p_{\min}$ and $p_{\max}$ proves Eq. (2.1); taking infima over strategies proves the value-one equivalence. A MERP strategy is perfect precisely when it satisfies the phase equation on every positive-weight clause. Deleting zero-weight clauses and merging repeated equal-target clauses therefore preserve both properties. If two clauses on the same question tuple have opposite targets and weights $p_0, p_1 > 0$, their combined loss is at least $\min\{p_0, p_1\}$, independently of the strategy. Such a pair therefore forces $\omega_{\mathrm{co}} < 1$ and also precludes a perfect MERP strategy. $\square$

For a MERP strategy, question j of player α is assigned the observable $M(\theta_{\alpha,j})$. Identifying the output bit a_α with the eigenvalue $(-1)^{a_\alpha}$, clause c is won with certainty exactly when

$$\sum_{\alpha=1}^{4} \theta_{\alpha, q_c^{(\alpha)}} = \pi b_c \pmod{2\pi}. \tag{2.2}$$

Thus perfect MERP satisfiability is an abelian phase-consistency problem.

2.2 Abelian and noncommutative obstructions

Let A be the clause-by-player-question incidence matrix of the reduced support:

$$A_{e,(\alpha,j)} = \begin{cases} 1, & q_e^{(\alpha)} = j, \\ 0, & q_e^{(\alpha)} \neq j. \end{cases}$$

We call the block of columns associated with a fixed player, and by extension the corresponding projection of a clause word, that player's *page*. Its integer relation lattice and its parity image are

$$K_E = \ker_{\mathbb{Z}}(A^{\mathsf{T}}), \tag{2.3}$$

$$L_E = \{y \bmod 2 : y \in K_E\} \subseteq \mathbb{F}_2^E. \tag{2.4}$$

An integer vector y is a *parity-permuted refutation specification* (PREF) if

$$A^{\mathsf{T}}y = 0, \qquad b^{\mathsf{T}}y = 1 \pmod 2. \tag{2.5}$$

The MERP–PREF duality gives

$$G \text{ has a perfect MERP strategy} \quad \Longleftrightarrow \quad b \in L_E^{\perp}. \tag{2.6}$$

The noncommutative obstruction retains the order of the clauses. A clause word $w = e_1 \cdots e_N$ is *balanced* if, for each player, the projected word in the question involutions freely reduces to the identity. Let $\pi(w) \in \mathbb{F}_2^E$ record the parity with which each clause occurs and set

$$R_E = \operatorname{span}_{\mathbb{F}_2}\{\pi(w) : w \text{ is balanced}\}. \tag{2.7}$$

A balanced word with $b^{\mathsf{T}}\pi(w) = 1$ is a *true refutation*.

For completeness, consider the standard game group generated by a central involution J and question involutions $x_{\alpha,j}$, with generators belonging to distinct players required to commute. Clause c is represented by

$$J^{b_c} \prod_{\alpha=1}^{4} x_{\alpha, q_c^{(\alpha)}}.$$

An ordered product of these clause elements equals J exactly when the corresponding clause word is a true refutation. The game-group criterion for finite binary XOR games therefore yields

$$\omega_{\text{co}}(G) = 1 \quad \Longleftrightarrow \quad \text{no true refutation} \quad \Longleftrightarrow \quad b \in R_E^{\perp}. \tag{2.8}$$

Here value one is attained in the commuting-operator model [WHKN19, WH23, WHK23]. Accordingly, a perfect commuting-operator strategy without a perfect MERP strategy occurs precisely when a PREF exists but no true refutation exists.

2.3 The Klein four-group witness

Write $V_4 = \{0, a, b, c\}$, with $a + b = c$, and index both the players and their questions by V_4. The eight clauses of the separating game are listed in Table 1.

Table 1: The Klein four-group game. The question tuple is ordered by the players $0, a, b, c$; all clauses have weight $1/8$, and the last column gives the target parity.

Clause	Question tuple	Target
E_0	$(0,0,0,0)$	0
E_a	(a,a,a,a)	0
E_b	(b,b,b,b)	0
E_c	(c,c,c,c)	0
O_0	$(0,a,b,c)$	1
O_a	$(a,0,c,b)$	0
O_b	$(b,c,0,a)$	0
O_c	$(c,b,a,0)$	0

Every player-question pair occurs once in the E-family and once in the O-family. Hence the coefficient assignment

$$E_g \longmapsto -1, \qquad O_g \longmapsto 1 \qquad (g \in V_4) \tag{2.9}$$

is an integer incidence relation. Because O_0 is the unique odd-target clause, its target-weighted parity is one, so the assignment is a PREF. Section 4.2 proves that no true refutation exists, irrespective of its length, clause order, or multiplicities. Equations (2.6) and (2.8) then give part (ii) of Theorem 1.1.

Subsection 4.2 proves the analogous all-length obstruction for every elementary abelian 2-group; the Klein game is the case of rank two.

3 Exact circuit lifting for four ternary pages

The MERP duality, comprising the phase characterization and the MERP–PREF alternative, follows from [WHKN19, Claim 5.29] and [WHKN19, Theorem 5.30]. The commuting-operator duality in Eq. (2.8) follows by translating the XOR game-group criterion of [WH23, Theorem 2.1] into the balanced-word notation of Subsection 2.2, where the corresponding clause group is defined in [WH23, Eq. (2.9)]. A broader Nullstellensatz formulation appears in [WHK23]. Using these two dualities, Theorem 1.1(i) reduces to the exactness of abelianization for four pages with at most three active questions, namely

$$L_E = R_E \qquad \text{for every } E \subseteq [3]^4. \tag{3.1}$$

The inclusion $R_E \subseteq L_E$ holds in general. The content of this section is the reverse inclusion for four-partite ternary supports, which requires an unordered integral incidence relation to be realized by a single clause ordering whose four page projections all freely reduce.

We prove a statement that retains the full integral multiplicity data. The *alternating vector* of a word $w = e_1 \cdots e_N$ is

$$y(w)_e = \sum_{\{t : e_t = e\}} (-1)^{t-1}.$$

A *primitive support-minimal circuit* is a nonzero vector $c \in \ker_{\mathbb{Z}}(A^{\top})$ such that no nonzero relation has support strictly contained in $\operatorname{supp} c$, and whose nonzero coordinates have greatest common divisor one. It is determined by its support up to sign.

Theorem 3.1 (Exact circuit lifting). *Let $E \subseteq [3]^4$ be a distinct clause support, let A be its clause-by-player-question incidence matrix, and let $c \in \ker_{\mathbb{Z}}(A^{\mathsf{T}})$ be a primitive support-minimal circuit. There exists a balanced word w_c that uses each clause e exactly $|c_e|$ times and satisfies*

$$y(w_c) = c \ \text{ or } \ -c. \tag{3.2}$$

In particular, $\pi(w_c) = c \bmod 2$.

Corollary 3.2. *For every distinct support $E \subseteq [3]^4$, one has $L_E = R_E$.*

The exact clause multiplicities are stronger than a parity realization for each primitive circuit. This strengthened conclusion makes the passage from integral circuit generation to the refutation space immediate. The proof has three ingredients. Support minimality localizes every circuit to at most ten clauses. Signed-occurrence matchings then convert the ordering problem into a linear-forest problem. Finally, ternary Hamming geometry excludes every configuration for which the required forest might fail to exist.

3.1 Algebraic localization

We record the algebraic reductions used to pass between the circuit-lifting theorem and the obstruction spaces, and then localize primitive circuits to bounded support.

Lemma 3.3. *For every balanced word $w = e_1 \cdots e_N$, its alternating vector satisfies $A^{\mathsf{T}} y(w) = 0$ and $y(w) \bmod 2 = \pi(w)$. Hence $R_E \subseteq L_E$.*

Proof. Fix a player and a complete free reduction of the corresponding projected word. Matched equal letters occupy positions $r < s$ whose intervening subword also reduces to the empty word. That subword has even length, so r and s have opposite parity. Their contributions cancel in the alternating sum for their common question. Summing over reduction pairs proves every coordinate of $A^{\mathsf{T}} y(w) = 0$. Modulo two, both signs equal 1, giving the parity identity. □

Thus only $L_E \subseteq R_E$ remains. The first reduction is genuinely integral: rational spanning of the relation space would not suffice for the parity conclusion.

Proposition 3.4. *The lattice K_E is generated over $\mathbb{Z}$ by its primitive support-minimal circuits.*

Proof. We use induction on $|\operatorname{supp} y|$, for $y \in K_E$. The assertion is immediate for $y = 0$. Otherwise choose a nonzero relation whose support is contained in $\operatorname{supp} y$ and is minimal among such supports. Dividing its coefficients by their greatest common divisor produces a primitive circuit c with $\operatorname{supp} c \subseteq \operatorname{supp} y$.

For $e \in \operatorname{supp} c$, the relation $c_e y - y_e c$ vanishes in coordinate e. If it is nonzero, its support is strictly smaller than $\operatorname{supp} y$, and hence it is an integer combination of primitive circuits by induction. Thus $c_e y = (c_e y - y_e c) + y_e c$ belongs to the integral circuit span in every case. Primitivity gives $\gcd\{c_e : e \in \operatorname{supp} c\} = 1$, so Bézout's identity supplies integers a_e with $\sum_{e \in \operatorname{supp} c} a_e c_e = 1$. Therefore

$$y = \sum_{e \in \operatorname{supp} c} a_e (c_e y)$$

also lies in the integer span of the primitive circuits. □

Lemma 3.5. *Let $c \in K_E$ be a circuit with support $S = \operatorname{supp} c$. Every player-question block that is active on S contains at least two clauses of S.*

Proof. For every player α and question j, the coordinate equation $(A^{\mathsf{T}} c)_{(\alpha,j)} = 0$ is the sum of c_e over clauses in that block. If its intersection with S were the singleton $\{e\}$, the equation would give $c_e = 0$, contradicting $e \in \operatorname{supp} c$. □

Proposition 3.6. *If c is a primitive circuit on S, then, for the restriction A_S of A to the rows indexed by S,*

$$\operatorname{rank}_{\mathbb{Q}} A_S = |S| - 1, \qquad |S| \le 2 + \sum_{\alpha=1}^{4} \left(|\{q_e^{(\alpha)} : e \in S\}| - 1 \right) \le 10.$$

At ten clauses all pages are ternary; at nine clauses the question-count profile is, up to permutation, $(3,3,3,2)$ or $(3,3,3,3)$.

Proof. If the rational relation space on S had dimension at least two, choose $z \in \mathbb{Q}^S$ independent of c, and clear denominators. Since every $c_e \neq 0$, some $c_e z - z_e c$ is nonzero and vanishes at e, contradicting minimal support after clearing denominators once more. Thus the nullity is one and the rank is $|S| - 1$.

For every page, its nonempty question-indicator columns sum to the same all-ones vector. Hence that vector, together with one fewer contrast than the number of active questions on each page, spans all incidence columns. This proves both the rank bound and the asserted question-count profiles. □

The rank cap shows that a primitive circuit has at most ten support clauses, although the coefficients $|c_e|$, and hence the length of an exact lift, remain unrestricted. It therefore remains to solve an ordering problem on at most ten distinct clause types. The next subsection constructs such orderings from signed-occurrence forests.

3.2 Exact lifts from signed-occurrence forests

We now isolate the noncommutative part of the proof. Fix a primitive circuit c. Its coefficients prescribe signed clause multiplicities but do not provide a common ordering in which all four page projections freely reduce. Replace clause e by $|c_e|$ occurrence vertices and put these vertices in parts P and N according to the sign of c_e. Summing the incidence equations over the questions of any one player gives $\sum_e c_e = 0$, so P and N have the same size. More precisely, every player-question block contains equally many vertices from the two parts.

We call c *liftable* if there is a balanced word using each clause e exactly $|c_e|$ times. Such a word is an *exact lift*; the following lemma shows that its position signs necessarily recover the circuit signs.

Lemma 3.7. *If a balanced word uses clause e exactly $|c_e|$ times, then its alternating vector is c or $-c$. Consequently, after replacing c by $-c$ if necessary, the positive and negative occurrences occupy the odd and even positions, respectively.*

Proof. By Lemma 3.3, the alternating vector is a rational relation on the circuit support. The relation space on that support is one-dimensional, so it equals λc. Since c is primitive and the alternating vector is integral, $\lambda \in \mathbb{Z}$; the coordinate bounds give $|\lambda| \le 1$. If $\lambda = 0$, then every count $|c_e|$ is even, because an occurrence count and its alternating sum have the same parity. This contradicts primitivity. Hence $\lambda = \pm 1$, and equality in every coordinate bound places all occurrences of one circuit sign on one position parity. $\square$

Lemma 3.8. *(i) A word on at most two letters with zero alternating sum for each letter freely reduces to the empty word. (ii) A finite bipartite linear forest with bipartition $P \sqcup N$ and $|P| = |N|$ has a total vertex order alternating between the parts and placing every forest edge consecutively.*

Proof. For (i), if the word has no adjacent equal letters, every nonempty reduced word on two letters alternates the letters. Its two letterwise alternating sums cannot both vanish. Thus an adjacent equal pair exists. Delete it and argue by induction; deleting two consecutive positions preserves the parity of every remaining position.

For (ii), traverse each path component from one endpoint to the other. A component of even order contains equally many vertices from P and N, and its traversal may start in either part. A component of odd order contains one extra vertex in the part containing its endpoints. The equality $|P| = |N|$ pairs the odd-order components with opposite excesses. Concatenating each such pair, and then the even-order components with suitable orientations, gives an alternating total order. Every path edge joins consecutive vertices in this order. $\square$

The matching problem is encoded by signatures. Select at most one question block on each ternary page and regard each selected block as a color. An occurrence vertex receives the set of colors of the selected blocks that contain it.

Definition 3.9 (Balanced signature system)**.** A balanced signature system consists of finite vertex sets P, N, a finite color set, and a signature (a subset of the colors) attached to each vertex, such that every color occurs equally often in P and N. A colorwise matching pairs positive and negative incidences of each color. If two endpoints share two colors, the same simple edge may serve both matchings; the forest condition refers to the union of the resulting simple edges.

Proposition 3.10. *If the selected question blocks admit positive-to-negative matchings whose simple union is a bipartite linear forest, then c has an exact lift.*

Proof. Order the occurrence vertices by Lemma 3.8(ii), with P and N alternating. Reading their clause labels gives a word with the prescribed multiplicities. Every matching edge joins consecutive positions. Consequently, on a page with a selected question, all occurrences of that question cancel in adjacent equal pairs. The same simple edge may carry two colors, in which case the corresponding pair cancels on both pages.

After these pairs are deleted from a fixed page projection, at most two question letters remain. Deletion takes place in pairs, so the position parities of the remaining letters do not change. Because the total order alternates between P and N, the alternating sum for every remaining question equals its coordinate in $A^{\mathsf{T}}c$, up to one common sign, and is therefore zero. Lemma 3.8(i) now freely reduces the residual page word. This holds on every page, so the clause word is balanced and is an exact lift. $\square$

Thus exact circuit lifting has been reduced to a matching statement: select question blocks and pair their signed occurrences so that the simple union has maximum degree two and no cycle. The next two propositions provide the matching statements needed below.

Proposition 3.11. *For at most three colors, every balanced signature system with signatures of size at most two has colorwise matchings whose simple union is a bipartite linear forest.*

Proof. Proceed by induction on the total number of color incidences. First match and remove opposite-sign vertices with identical nonempty signatures, using one simple edge for both colors of a two-color signature. Record these isolated edges and restore them after treating the residual system. No nonempty signature then occurs on both signs in that system.

If a singleton $\{a\}$ remains, match its a-incidence to any opposite-sign endpoint incident with a, delete a from both signatures, and apply induction. The singleton endpoint is isolated in the residual system; restoring the edge therefore creates neither a cycle nor degree greater than two.

It remains to exclude an active residual system consisting only of two-element signatures. For $\sigma \in \{ab, ac, bc\}$, let δ_σ be the multiplicity of signature σ in P minus its multiplicity in N. Color balance says

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} \delta_{ab} \\ \delta_{ac} \\ \delta_{bc} \end{pmatrix} = 0.$$

The matrix has determinant -2, so every $\delta_\sigma = 0$. Since no nonempty signature occurs on both signs in the residual system, all three multiplicities must vanish, a contradiction. The cases with fewer active colors are contained in the same argument. $\square$

Proposition 3.12. *For four colors, a balanced signature system of signature size at most two has a linear-forest matching whenever either sign contains a singleton.*

Proof. Induct on total color incidence. Let a positive vertex have signature $\{a\}$. If a negative endpoint has signature $\{a, b\}$, delete a from both endpoints. The positive endpoint becomes empty and the negative endpoint becomes the singleton $\{b\}$; apply induction and restore the edge incident with the now-isolated positive endpoint.

Otherwise every negative endpoint incident with a has signature $\{a\}$. Delete one positive-negative singleton pair. If another singleton remains, recurse. If not, no remaining negative endpoint uses a, and color balance says no positive endpoint uses it either. The residual system uses only three colors, so Proposition 3.11 applies. Restore the isolated pair. The negative-singleton case is symmetric. $\square$

The support bound and the three-color forest already establish liftability for every circuit that does not use three questions on all four pages.

Proposition 3.13. *Let c be a primitive circuit with $|\operatorname{supp} c| \leq 10$. If at most three pages use three questions on* $\operatorname{supp} c$*, then c is liftable.*

Proof. If at most two pages are ternary, select one question block on each of them. There are at most two colors, so every signature has size at most two and Propositions 3.11 and 3.10 apply.

Suppose exactly three pages are ternary. The projection of at most ten clauses onto these pages omits a point of $[3]^3$. Select the three question blocks defining such an omitted point. No occurrence has all three colors, so again every signature has size at most two. The same two propositions give an exact lift. Pages using at most two questions require no selected block and are handled by Lemma 3.8(i). $\square$

3.3 Hamming geometry of a nonliftable circuit

By Proposition 3.13, a nonliftable primitive circuit must use all three questions on every page. We identify its support with a subset $S \subset [3]^4$ of the ternary Hamming cube and use a point outside the radius-one neighborhood of S to select one block on each page.

Set

$$U(S) = \{u \in [3]^4 : \mathrm{d}(u, S) \geq 2\},$$

and for $u \in U(S)$ write $a_u = |\{q \in S : \mathrm{d}(u, q) = 2\}|$. We call the points of $U(S)$ *safe centers*. At a safe center u, select the four question blocks specified by the coordinates of u. The signature of a clause q is its set of agreements with u, and hence has size $4 - \mathrm{d}(u, q) \leq 2$.

If $\mathrm{d}(u, q) = 3$, this signature is a singleton. Propositions 3.12 and 3.10 would then lift the circuit. Thus nonliftability forces the even-layer condition

$$\mathrm{d}(u, q) \in \{2, 4\} \qquad (u \in U(S),\ q \in S). \tag{3.3}$$

A safe center is therefore a certificate of liftability unless the support has this rigid even-layer form.

Lemma 3.14. *For a nonliftable circuit with support $S \subset [3]^4$, using all three questions on each page and having at most ten clauses, distinct safe centers have distance at least three and $|U(S)| \leq 6$.*

Proof. Suppose first that safe centers u, u' differ only on page a, and let r_a be the third question value on that page. Comparing the parities in Eq. (3.3) for u and u' forces every clause to use r_a, contradicting the assumption that page a is ternary.

If they differ on pages a, b, with third values r_a, r_b, parity comparison gives

$$\mathbf{1}_{\{q^{(a)} \neq r_a\}} = \mathbf{1}_{\{q^{(b)} \neq r_b\}}$$

for every clause. Thus the complete r_a-block on page a equals the complete r_b-block on page b. On either page, the two other active question blocks have size at least two by Lemma 3.5. Since $|S| \leq 10$, the common block has size at most six. Its projection onto the remaining two pages omits one of their nine joint cells. Regard the common occurrence block as one color and select the two question blocks defining the omitted cell as the other colors. No occurrence has all three colors, so Proposition 3.11 supplies a linear forest. The matching of the common block can be used simultaneously on pages a and b, because the two selected blocks are identical as multisets of signed occurrences. Proposition 3.10 then gives an exact lift. This contradiction proves that distinct safe centers are separated by distance at least three.

For the packing bound, fix a clause and relabel it as 0. By Eq. (3.3), every safe center lies in Hamming layer two or four about this clause. The support coordinates of a layer-two center define an edge on the four coordinate positions. Two centers cannot define the same edge, since their mutual distance would be at most two. Moreover, two incident edges must carry different nonzero labels at their common coordinate; otherwise the corresponding centers again have distance two. There are only two nonzero labels, so this simple graph has maximum degree two and hence at most four edges.

Layer-four centers belong to $\{1, 2\}^4$. This binary four-cube contains no three points at mutual distance at least three: relative to one point, the change sets of two others both have size at least three and hence have symmetric difference at most two. There are therefore at most two layer-four centers, and $|U(S)| \leq 4 + 2 = 6$. $\square$

Lemma 3.15. *Let c be a nonliftable primitive circuit with support $S \subset [3]^4$, using three questions on all four pages and satisfying $|S| \le 10$. The total excess multiplicity of the radius-one balls centered at S—the sum, over covered points, of their multiplicity minus one—is*

$$9|S| - 81 + |U(S)|.$$

Moreover,

$$a_u \ge 4 \qquad (u \in U(S)), \tag{3.4}$$

$$2 \sum_{u \in U(S)} (a_u - 4) \le 9|S| - 81 + |U(S)|. \tag{3.5}$$

Proof. There are $9|S|$ incidences between the radius-one balls centered at S and the 81-point cube. Exactly $81 - |U(S)|$ points are covered; subtracting one from every covered multiplicity gives the stated total excess.

Fix $u \in U(S)$. Each of its eight neighbors is covered: otherwise that neighbor would also belong to $U(S)$, contrary to the separation in Lemma 3.14. The radius-one neighborhoods of distinct safe centers are disjoint for the same reason. By Eq. (3.3), a support clause is at distance two or four from u. A distance-two clause covers exactly two of the eight neighbors of u, whereas a distance-four clause covers none. Hence the excess over these eight covered points is

$$2a_u - 8 = 2(a_u - 4) \ge 0,$$

which proves $a_u \ge 4$. The disjoint safe-center neighborhoods contribute disjoint parts of the total excess computed above; summing their local excesses proves Eq. (3.5). □

3.4 Exclusion of the extremal configurations

Only circuits on four ternary pages remain. By Proposition 3.6, their support size is at most ten. We now exclude the possible nonliftable configurations. The Hamming-ball excess is already decisive up to eight points. At nine points, a covering configuration would be a perfect packing. At ten points, the noncovering and covering cases are governed, respectively, by the layer-two fibers of a safe center and by the cosets of the ternary Hamming code.

Proposition 3.16. *Every primitive circuit on at most eight distinct clauses lifts.*

Proof. Circuits with at most three ternary pages lift by Proposition 3.13. Suppose that all four pages are ternary and, towards a contradiction, that a circuit with support S is nonliftable. Lemmas 3.14 and 3.15 give

$$0 \le 2 \sum_{u \in U(S)} (a_u - 4) \le 9|S| - 81 + |U(S)| \le 72 - 81 + 6 = -3.$$

The same calculation includes the case $U(S) = \varnothing$, for which the left side is zero. □

We next treat nine-point supports.

Lemma 3.17. *If nine radius-one balls centered at distinct points of $[3]^4$ cover the cube, the clause-question incidence matrix A_S of their centers has* $\operatorname{rank}_{\mathbb{Q}} A_S = 9$.

Proof. The nine balls have total size 81, so a cover makes them disjoint and their centers pairwise distance at least three. Projection of the centers onto any two coordinates is injective and hence bijective onto $[3]^2$. It follows that, on the nine centers, the two-dimensional zero-mean contrast spaces from distinct coordinates are mutually orthogonal; each is also orthogonal to the constants. The constant direction together with the four contrast spaces therefore has dimension $1 + 4 \cdot 2 = 9$ in the incidence column span. Hence the row rank is nine. □

Proposition 3.18. *Every nine-clause primitive circuit lifts.*

Proof. Let S be the circuit support. By Proposition 3.6, the question-count profile is, up to permutation, $(3, 3, 3, 2)$ or $(3, 3, 3, 3)$. The first case lifts by Proposition 3.13. In the second case, assume nonliftability. If $U(S) = \varnothing$, the nine radius-one balls cover the cube; Lemma 3.17 gives rank $A_S = 9$, whereas a nine-point circuit has rank eight by Proposition 3.6. Hence $U(S) \neq \varnothing$.

For nine support points, Eq. (3.5) becomes

$$2 \sum_{u \in U(S)} (a_u - 4) \leq |U(S)|.$$

Since every $a_u \geq 4$, some safe center satisfies $a_u = 4$; otherwise the left side would be at least $2|U(S)|$.

The Hamming layer two about u has $\binom{4}{2} 2^2 = 24$ points. The ball of a distance-two clause meets this layer in exactly three points, whereas a distance-four ball misses it. The four distance-two clauses therefore cover at most twelve layer-two points. Every remaining point in that layer is uncovered and hence is a safe center. This produces at least twelve safe centers in addition to u, contrary to Lemma 3.14. □

It remains to consider ten-point supports. We first treat the noncovering case, in which a safe center exists.

Proposition 3.19. *A ten-clause primitive circuit with support S and $U(S) \neq \varnothing$ lifts.*

Proof. Assume nonliftability. Lemma 3.14 gives $1 \leq |U(S)| \leq 6$. If $a_u \leq 6$ for some u, the radius-one balls of the distance-two clauses cover at most $3a_u$ of the 24 points in layer two about u; the distance-four balls cover none. Thus at least $24 - 3a_u \geq 6$ further points belong to $U(S)$, which together with u contradicts $|U(S)| \leq 6$. Hence $a_u \geq 7$ for every $u \in U(S)$. Equation (3.5) now gives

$$6|U(S)| \leq 9 + |U(S)|,$$

so $|U(S)| = 1$.

Relabel the unique center as 0. Its 24 layer-two points split into six fibers according to their two-coordinate support, with four choices of nonzero labels in each fiber. Only a distance-two clause with the same coordinate support can cover a point in a given fiber, and the ball of one such clause covers three of its four points. Since 0 is the unique safe center, every layer-two point must be covered. Each of the six fibers therefore requires at least two clauses, for a total of at least twelve, contrary to $|S| = 10$. □

We finally treat the case in which the ten radius-one balls cover the cube.

Proposition 3.20. *Ten distinct radius-one balls covering $[3]^4$ cannot have centers that support a primitive circuit.*

Proof. Let S be the set of ten centers and consider the ternary $[4, 2, 3]_3$ Hamming code

$$C = \{(a, b, a + b, a + 2b) : a, b \in \mathbb{F}_3\}.$$

Any two zero coordinates in a codeword force $a = b = 0$; hence every nonzero codeword has weight at least three. Since $(1, 0, 1, 1) \in C$, the displayed code has minimum distance three. Its nine cosets partition $\mathbb{F}_3^4$ into codes of size nine.

Every radius-one ball meets each coset exactly once. Indeed, the zero error and the eight nonzero errors supported on one coordinate represent distinct cosets of C, because the difference of two such errors has weight at most two. Since $\mathbb{F}_3^4/C$ has nine elements, these errors form a complete set of coset representatives; translation by a ball center proves the claim.

Ten balls contribute ten incidences to each nine-point coset. If they cover the cube, every point has positive multiplicity, so each coset contains exactly one double point and eight single points. Across the nine cosets there are exactly nine double points and no point of higher multiplicity. If n_d denotes the number of unordered pairs of centers at distance d, two radius-one balls meet in three points at center distance one, in two points at distance two, and not at all at larger distances. Counting the nine double points gives

$$3n_1 + 2n_2 = 9. \tag{3.6}$$

Suppose S supports a primitive circuit. Every page is ternary and, by Lemma 3.5, every question block has size at least two. Among partitions of ten into three parts of size at least two, $(4, 3, 3)$ minimizes the number of within-block pairs, giving

$$\binom{4}{2} + \binom{3}{2} + \binom{3}{2} = 12.$$

Summed over four pages, the number of coordinate agreements satisfies

$$3n_1 + 2n_2 + n_3 \geq 48.$$

Together with $n_1 + n_2 + n_3 + n_4 = 45$, this implies

$$2n_1 + n_2 - n_4 \geq 3.$$

With e_0, e_1, e_2 denoting the standard basis of $\mathbb{R}^3$, define

$$\Phi(q) = \left(\tfrac{1}{\sqrt{3}}, e_{q_1} - \tfrac{1}{3}\mathbf{1}, e_{q_2} - \tfrac{1}{3}\mathbf{1}, e_{q_3} - \tfrac{1}{3}\mathbf{1}, e_{q_4} - \tfrac{1}{3}\mathbf{1}\right).$$

The Gram matrix of the ten feature vectors satisfies

$$\mathcal{G}_{ij} = \langle \Phi(q_i), \Phi(q_j) \rangle = 3 - \mathrm{d}(q_i, q_j).$$

Centered and raw question indicators have the same column span after the constant column is included. Proposition 3.6 therefore gives $\operatorname{rank} \mathcal{G} = \operatorname{rank} A_S = 9$, and $\mathcal{G}$ and $A_S^\top$ have the same kernel on $\mathbb{R}^S$.

The off-diagonal nonzero graph of $\mathcal{G}$ joins pairs at distances one, two, and four. It must be connected. Otherwise $\mathcal{G}$ is block diagonal, and restricting the full-support circuit vector to each connected component gives a nonzero kernel vector on that component. Two components would yield two independent kernel vectors, contrary to the one-dimensional circuit kernel. A connected graph on ten vertices has at least nine edges, whereas Eq. (3.6) gives

$$n_1 + n_2 + n_4 = 9 - (2n_1 + n_2 - n_4) \le 6.$$

This contradiction proves the claim. $\square$

Proof of Theorem 3.1. Let c be a primitive circuit with support S. Proposition 3.6 gives $|S| \le 10$. If at most three pages are ternary, Proposition 3.13 supplies an exact lift. If all four pages are ternary, Propositions 3.16 and 3.18 handle support sizes at most nine. For support size ten, either $U(S) \neq \varnothing$, in which case Proposition 3.19 applies, or the ten radius-one balls cover the cube, which Proposition 3.20 shows is incompatible with a primitive circuit. Thus every primitive circuit has an exact lift. Its alternating vector is c or $-c$ by Lemma 3.7, proving Eq. (3.2). $\square$

Proof of Corollary 3.2. Proposition 3.4 writes every element of K_E as an integer combination of primitive circuits. Reducing this identity modulo two shows that their parity vectors span L_E over $\mathbb{F}_2$. Each such parity vector equals $\pi(w_c)$ for a balanced word, and hence belongs to R_E. Therefore $L_E \subseteq R_E$. The reverse inclusion is Lemma 3.3, so $L_E = R_E$. Notice that this identity depends only on the support E, not on the target vector b. $\square$

Proof of Theorem 1.1(i). If one question tuple occurs with both targets, the commuting-operator value is strictly below one and no perfect MERP strategy exists, as observed in Proposition 2.1. Otherwise reduce to a distinct positive-weight support. Proposition 2.1 permits the uniform distribution, and the active-question hypothesis identifies the support with a subset $E \subseteq [3]^4$. Corollary 3.2 gives $L_E = R_E$. The dualities in Eqs. (2.6) and (2.8) now yield

$$\omega_{\text{co}}(G) = 1 \iff b \in R_E^{\perp} \iff b \in L_E^{\perp} \iff G \text{ admits a perfect MERP strategy.}$$

$\square$

4 Sharpness at four questions

Part (ii) of Theorem 1.1 requires more than the PREF exhibited in Subsection 2.3: one must also exclude every true refutation. We separate this task into two logically distinct levels. First, we classify the shortest natural candidates—words using every Cayley clause exactly once—for an arbitrary finite group. We then specialize to elementary abelian 2-groups and use the Magnus expansion to rule out refutations with arbitrary order and multiplicity. The second step is essential; the first alone cannot establish commuting-operator value one.

4.1 Each-clause-once refutations in Cayley games

Before addressing refutations of arbitrary length, we isolate the shortest natural candidate: a word in which every Cayley clause occurs exactly once. For each player, free reduction of such a word is equivalent to a noncrossing matching of equal questions. The group translations produce a family of matchings that must be noncrossing in one common linear order. Although simultaneous noncrossing matchings are related to partitioned book embeddings [ADHL18], the regular group action makes this family completely classifiable.

Let H be a finite group of order n. The associated Cayley game has players and question labels indexed by H, clauses

$$E_g = (g)_{\alpha \in H}, \qquad O_g = (g\alpha)_{\alpha \in H},$$

and the unique odd target on O_1. For $h \in H$, let m_h denote the perfect matching between the two clause copies defined by $E_x \leftrightarrow O_{xh}$.

Theorem 4.1. *The family $\{m_h : h \in H\}$ admits a common noncrossing linear order if and only if H is cyclic. For $n \geq 2$, this is equivalent to the existence of a true refutation that uses every clause of the associated Cayley game exactly once.*

This theorem explains the failure of an each-clause-once refutation for the Klein group, but it is not by itself an all-length obstruction. The latter requires the Magnus argument of Subsection 4.2.

For $n = 1$, the matching classification is immediate. The two indexed clauses then have the same question tuple and opposite targets, so that presentation is not reduced. We henceforth assume $n \geq 2$ when referring to the game.

We begin with the elementary word criterion underlying the matching formulation.

Lemma 4.2. *Let W be a word of length $2n$ in which each of n involutive letters occurs exactly twice. The word W freely reduces to the identity if and only if the matching that joins equal letters is noncrossing in the linear order of the word.*

Proof. For a noncrossing matching, an innermost chord has consecutive endpoints. Deleting that equal adjacent pair preserves noncrossing, and induction reduces the word to the identity. Conversely, suppose that W freely reduces. If W is nonempty, the first deletion in a free reduction removes an adjacent equal pair. Its chord crosses no other chord. Delete the pair; the shorter word freely reduces and its matching is noncrossing by induction. Reinserting the consecutive chord preserves noncrossing. $\square$

In a length-$2n$ Cayley word using every clause once, player α pairs E_x with $O_{x\alpha^{-1}}$, because the question received on O_g is $g\alpha$. As inversion permutes H, Lemma 4.2 identifies a true refutation of this form with a common noncrossing order for the matching family $\{m_h : h \in H\}$.

We first prove that a common noncrossing order forces H to be cyclic. Suppose that such an order exists. If its first point is an O-point, interchange the two copies E and O. This sends m_h to $m_{h^{-1}}$ and preserves the matching family. A common left translation of both copies then makes the first point E_1. Thus no generality is lost by assuming that the order begins with E_1.

Lemma 4.3. *For every $x \neq 1$ and $g \in H$,*

$$E_x < O_g \iff O_{xg} < O_g. \tag{4.1}$$

Proof. In m_g, the chord incident with E_1 is (E_1, O_g), while the chord incident with E_x is (E_x, O_{xg}). Since E_1 is the first point and the two chords do not cross, the endpoints E_x and O_{xg} lie on the same side of O_g. This is Eq. (4.1). $\square$

List the O-points in their order of appearance:

$$O_{o_1} < O_{o_2} < \cdots < O_{o_n},$$

and define $S_j = \{x \in H : E_x < O_{o_j}\}$. Applying Eq. (4.1) with $g = o_j$ gives

$$S_j = \{1\} \cup \{o_i o_j^{-1} : 1 \le i < j\}. \tag{4.2}$$

The displayed elements are distinct, and none of the latter elements equals the identity. Hence $|S_j| = j$ and

$$S_1 \subsetneq S_2 \subsetneq \cdots \subsetneq S_n.$$

For $1 \le j < n$, set $d_j = o_j o_{j+1}^{-1}$. Eq. (4.2) yields

$$S_{j+1} = \{1\} \cup (S_j \setminus \{1\}) d_j \cup \{d_j\}. \tag{4.3}$$

Lemma 4.4. *There is an element $d \in H$ such that*

$$d_j = d \quad (1 \le j < n), \qquad S_j = \{1, d, \dots, d^{j-1}\} \quad (1 \le j \le n).$$

Proof. Set $d = d_1$, so $S_2 = \{1, d\}$. Suppose inductively that $S_j = \{1, d, \dots, d^{j-1}\}$. The recurrence gives

$$S_{j+1} = \{1, d_j, dd_j, \dots, d^{j-1} d_j\}.$$

Because $d \in S_j \subset S_{j+1}$, either $d = d_j$, or $d = d^k d_j$ for some $1 \le k \le j-1$. The case $k = 1$ would give $d_j = 1$, contrary to $o_j \ne o_{j+1}$. If $k \ge 2$, then

$$S_{j+1} = \{1\} \cup \{d^{1-k}, d^{2-k}, \dots, d^{j-k}\}.$$

The exponent interval contains zero, so the identity occurs both outside and inside the second set; consequently $|S_{j+1}| \le j$, contrary to $|S_{j+1}| = j+1$. Therefore $d_j = d$, and Eq. (4.3) gives $S_{j+1} = \{1, d, \dots, d^j\}$. $\square$

Since $S_n = H$ has n elements, Lemma 4.4 implies that d has order n and $H = \langle d \rangle$. Moreover, the new E-point entering between O_{o_j} and $O_{o_{j+1}}$ is E_{d^j}, while $o_{j+1} = o_j d^{-1}$. Thus the common order is forced to alternate between the two copies.

It remains to prove that cyclicity is sufficient. Let $H = \langle d \rangle$ have order n and consider the zero-indexed order

$$E_1, O_1, E_d, O_{d^{-1}}, \dots, E_{d^{n-1}}, O_{d^{-(n-1)}}. \tag{4.4}$$

Fix $h = d^k$. In the matching m_h, the point E_{d^i} at position $2i$ is paired with the O-point at position $2j+1$, where j is the representative of $-k-i \pmod{n}$ in $\{0, \dots, n-1\}$. To compare two chords, take $i_1 < i_2$, let j_s be the corresponding representative for i_s, and put $\Delta = i_2 - i_1$.

If $j_2 = j_1 - \Delta$, then $j_2 < j_1$. A crossing would require either

$$2i_1 < 2j_2 + 1 < 2j_1 + 1 < 2i_2$$

or

$$2j_2 + 1 < 2i_1 < 2i_2 < 2j_1 + 1.$$

The first set of inequalities gives $j_2 \ge i_1$ and $j_1 < i_2$; substitution of $j_2 = j_1 - \Delta$ instead gives $j_1 \ge i_2$. The second gives $j_2 < i_1$ and $j_1 \ge i_2$; substitution instead gives $j_1 < i_2$. Both alternatives are impossible.

If the index wraps, then $j_2 = j_1 + n - \Delta > j_1$. Because $j_2 < n$, one has $\Delta > j_1$, so $2i_2 > 2j_1 + 1$. Moreover $\Delta \le n - 1 - i_1$ gives $j_2 \ge i_1 + 1$, and hence $2j_2 + 1 > 2i_1$. The two alternating endpoint orders that could produce a crossing are thereby excluded. Thus m_h is noncrossing for every $h \in H$.

By Lemma 4.2, the order Eq. (4.4) gives a true refutation. The target parity is odd because O_1 occurs exactly once. This proves Theorem 4.1.

4.2 All-length obstruction for elementary abelian Cayley games

This subsection proves the all-length assertion used in part (ii) of Theorem 1.1. Let $H = \mathbb{F}_2^r$, $r \ge 2$, and assume that a true refutation of length L exists. Write the clause in position p as (g_p, m_p), where $m_p = 0$ for E_{g_p} and $m_p = 1$ for O_{g_p}. Player α receives question $g_p + m_p\alpha$. We first rewrite each player projection in a free basis for the even subgroup. Degree one of the Magnus expansion fixes all alternating clause sums in terms of a single odd integer. Degree two then supplies pair-count identities, and an integral certificate contradicts that oddness.

We begin with the required normal form for the even subgroup.

Let $F = *_{q\in H}\mathbb{Z}_2$ be generated by question involutions x_q, and let F^{e} be its even subgroup.

Lemma 4.5. *For $q \neq 0$, set $t_q = x_q x_0$, and put $t_0 = 1$. The elements $\{t_q : q \neq 0\}$ freely generate F^{e}. Every even word satisfies*

$$x_{a_1} \cdots x_{a_{2N}} = \prod_{i=1}^{N} t_{a_{2i-1}} t_{a_{2i}}^{-1}.$$

Proof. The homomorphism from F to $\mathbb{Z}_2$ that sends every x_q to the nontrivial element has kernel F^{e}, with Schreier transversal $\{1, x_0\}$. The Reidemeister–Schreier generators associated with x_q, $q \neq 0$, are

$$x_q x_0 = t_q, \qquad x_0 x_q = t_q^{-1}.$$

The generator associated with x_0 is trivial, and rewriting the relations $x_q^2 = 1$ produces only the tautological relations $t_q t_q^{-1} = 1$. The subgroup is therefore free on the t_q's [MKS76]. Finally, $t_q t_s^{-1} = x_q x_s$; multiplying this identity over consecutive pairs proves the displayed normal form. □

Consequently, the projection of player α reduces to the identity if and only if

$$\prod_{i=1}^{L/2} X_{g_{2i-1}+m_{2i-1}\alpha} X^{-1}_{g_{2i}+m_{2i}\alpha} = 1 \tag{4.5}$$

in the free group generated by X_q, $q \neq 0$, with $X_0 = 1$.

We use the degree-one and class-two terms of the Magnus expansion [Mag37, MKS76].

Lemma 4.6. *Let $W = X_{a_1}^{\varepsilon_1} \cdots X_{a_N}^{\varepsilon_N}$ be a word in a free group, with $\varepsilon_p \in \{\pm 1\}$, and let $[P]$ denote the indicator of a statement P. If $W = 1$, then*

$$\sum_p \varepsilon_p [a_p = i] = 0, \tag{4.6}$$

$$\sum_{p<q} \varepsilon_p \varepsilon_q [a_p = i][a_q = j] = 0 \qquad (i \neq j). \tag{4.7}$$

Proof. Apply the Magnus homomorphism into noncommutative formal power series, truncated above degree two:

$$X_i \longmapsto 1 + u_i, \qquad X_i^{-1} \longmapsto 1 - u_i + u_i^2.$$

The coefficient of u_i is the left-hand side of Eq. (4.6). For $i \neq j$, the coefficient of the ordered monomial $u_i u_j$ receives no contribution from the quadratic term of a single inverse and is therefore exactly the left-hand side of Eq. (4.7). If $W = 1$, every positive-degree coefficient vanishes, proving both identities. □

We first extract the degree-one consequences.

Let $s_p = 1$ for odd p and $s_p = -1$ for even p, and define

$$V_g = \sum_{\substack{p:g_p=g \\ m_p=0}} s_p, \qquad W_g = \sum_{\substack{p:g_p=g \\ m_p=1}} s_p.$$

The exponent of X_j in the word on the left of Eq. (4.5) is

$$V_j + W_{j+\alpha}.$$

Eq. (4.6), for $j \neq 0$, therefore implies

$$W_g = -v \quad (g \in H), \qquad V_j = v \quad (j \neq 0)$$

for one integer v. Since L is even, $\sum_p s_p = 0$, and summing all V_g and W_g gives $V_0 = v$.

Only O_0 has odd target parity. Modulo two,

$$W_0 = \sum_{p:O_0 \text{ occurs at } p} s_p \equiv \#\{p : O_0 \text{ occurs at } p\}.$$

The parity condition of a true refutation thus forces v to be odd.

We now turn to the degree-two constraints.

For distinct nonzero $i, j \in H$, define the signed pair counts

$$R_{ij} = \sum_{\substack{p<q \\ E_i \text{ at } p,\ E_j \text{ at } q}} s_p s_q, \qquad P_{ig} = \sum_{\substack{p<q \\ E_i \text{ at } p,\ O_g \text{ at } q}} s_p s_q,$$

$$Q_{gj} = \sum_{\substack{p<q \\ O_g \text{ at } p,\ E_j \text{ at } q}} s_p s_q, \qquad S_{gh} = \sum_{\substack{p<q \\ O_g \text{ at } p,\ O_h \text{ at } q}} s_p s_q.$$

For fixed nonzero $i \neq j$, the four possible combinations of clause families contribute the terms R, P, Q, S, respectively. Applying Eq. (4.7) to player c gives the system

$$P_{i,j+c} + Q_{i+c,j} + S_{i+c,j+c} = -R_{ij} \qquad (i \neq j,\ c \in H). \tag{4.8}$$

The constant term R_{ij} cannot be separated from the point masses in this equation because the constant function is their sum.

Splitting products of the degree-one sums into the regions $p < q$ and $q < p$ gives

$$R_{ij} + R_{ji} = v^2 \qquad (i \neq j), \tag{4.9}$$

$$P_{ig} + Q_{gi} = -v^2 \qquad (i \neq 0), \tag{4.10}$$
$$S_{gh} + S_{hg} = v^2 \qquad (g \neq h). \tag{4.11}$$

For example, the first identity is the equality $V_i V_j = v^2$ decomposed according to the order of the two occurrences; the other two are $V_i W_g = -v^2$ and $W_g W_h = v^2$.

It remains to combine these identities into a parity contradiction.

Identify a four-element subgroup of H with $\{0,1,2,3\} \cong V_4$, using bitwise XOR for its addition. Every equation whose indices lie in this subgroup is a member of the full system. For the six ordered pairs $i \neq j$ in $\{1,2,3\}$, take the four equations (4.8), indexed by $c = 0,1,2,3$, with the coefficients

(i,j)	$c=0$	$c=1$	$c=2$	$c=3$
$(1,2)$	5	-3	5	-3
$(1,3)$	3	3	-5	-5
$(2,1)$	3	-5	3	-5
$(2,3)$	-3	5	-3	5
$(3,1)$	-3	5	-3	-3
$(3,2)$	3	3	-5	3

To this weighted sum, add four times Eq. (4.9) for $(i,j) = (1,2)$ and $(1,3)$, subtract four times its $(2,3)$ instance, add eight times Eq. (4.10) for $(i,g) = (1,1)$, and subtract eight times its $(1,2)$ instance. Collection by variable family gives

family	surviving contribution
R	$8R_{12}$
P	0
Q	$8(Q_{13} - Q_{23} - Q_{31} + Q_{32})$
S	$8(-S_{31} + S_{32})$
right-hand side	$4v^2$

and therefore
$$8\bigl(R_{12} + Q_{13} - Q_{23} - Q_{31} + Q_{32} - S_{31} + S_{32}\bigr) = 4v^2.$$

The expression in parentheses is an integer, so v^2 is even. This contradicts the oddness of v and proves that no true refutation exists.

Finally, Eq. (2.9) is a PREF because every player-question incidence occurs once in each clause family, whereas only O_0 has odd target parity. MERP–PREF and strategy–refutation duality then give the two strategy consequences in part (ii) of Theorem 1.1. The conclusion is purely algebraic and does not rely on a bounded search over refutation words.

5 Discussion

Theorem 1.1 determines the sharp local-question threshold for four-player XOR games: with at most three active questions per player, every game of commuting-operator value one admits a perfect MERP realization, whereas four questions already allow failure. Proposition 2.1 shows that this threshold depends only on the active incidence pattern and target vector, not on the positive clause weights.

The proof identifies the structural change at the threshold. Below it, the integer relation lattice is generated by primitive circuits supported on at most ten clauses, and the forest and Hamming-geometric arguments lift each such circuit to a balanced word. If the original relation pairs oddly with the target vector, this lift gives a true refutation. At four questions, the Klein translation pattern has an abelian obstruction, but its degree-one and degree-two Magnus constraints exclude refutations of every length. The separation is therefore not a consequence of bounded enumeration: both directions are controlled by uniform structural arguments.

The Cayley analysis also distinguishes a local obstruction from an all-length one. Theorem 4.1 shows that a Cayley game has an each-clause-once refutation exactly when its indexing group is cyclic. For the elementary abelian family, Subsection 4.2 strengthens noncyclicity to the absence of refutations with arbitrary multiplicities.

Open problems. The first question is whether the Klein game admits a perfect finite-dimensional strategy. A positive answer would place the first escape from the GHZ-phase model already in finite dimension; a negative answer would separate finite-dimensional tensor-product from commuting-operator perfectness.

A second problem is to classify the supports $E \subseteq [4]^4$ for which $R_E \subsetneq L_E$. In particular, what is the minimum number of clauses, and are all minimal examples equivalent, up to relabeling, to a Klein-type translation pattern?

One may also ask for a quantitative version below the threshold. If every player has at most three active questions and $\omega_{\mathrm{co}}(G) \geq 1 - \varepsilon$, it is unclear whether there exist a GHZ-equatorial strategy of value at least $1 - f(\varepsilon)$ for an explicit rate $f(\varepsilon) \to 0$, preferably uniform over the clause weights. An robust game-algebra methods may provide a useful framework [Zha24].

Finally, the all-length obstruction should be extended beyond elementary abelian 2-groups. For which finite groups can repeated Cayley clauses form a true refutation, and when do degree-two Magnus coefficients suffice to rule this out, remains open.

Acknowledgments

This work was supported by the Guangdong Provincial Quantum Science Strategic Initiative under Grants No. GDZX2503001 and No. GDZX2403001.

Use of AI-assisted tools. Exploratory computation and the generalization of proofs in the present work were supported by a research harness system currently under development by QudeLeap Research and powered by DeepSeek V4 Flash. The authors independently formulated the research questions, established the mathematical conventions, and rigorously verified every statement and proof included in the manuscript. The authors take full responsibility for the final text and all mathematical claims contained therein.

References

[ADHL18] Hugo A. Akitaya, Erik D. Demaine, Adam Hesterberg, and Quanquan C. Liu. Upward partitioned book embeddings. In *Graph Drawing and Network Visualization*, volume 10692 of *Lecture Notes in Computer Science*, pages 210–223. Springer, 2018.

[Bel64] John S. Bell. On the Einstein–Podolsky–Rosen paradox. *Physics Physique Fizika*, 1(3):195–200, 1964.

[BV13] Jop Briët and Thomas Vidick. Explicit lower and upper bounds on the entangled value of multiplayer XOR games. *Communications in Mathematical Physics*, 321(1):181–207, 2013.

[CHSH69] John F. Clauser, Michael A. Horne, Abner Shimony, and Richard A. Holt. Proposed experiment to test local hidden-variable theories. *Physical Review Letters*, 23(15):880–884, 1969.

[CHTW04] Richard Cleve, Peter Høyer, Benjamin Toner, and John Watrous. Consequences and limits of nonlocal strategies. In *Proceedings of the 19th IEEE Conference on Computational Complexity*, pages 236–249, 2004.

[CLS17] Richard Cleve, Li Liu, and William Slofstra. Perfect commuting-operator strategies for linear system games. *Journal of Mathematical Physics*, 58(1):012202, 2017.

[dMU21] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction—CADE 28*, volume 12699 of *Lecture Notes in Computer Science*, pages 625–635. Springer, 2021.

[GHSZ90] Daniel M. Greenberger, Michael A. Horne, Abner Shimony, and Anton Zeilinger. Bell's theorem without inequalities. *American Journal of Physics*, 58(12):1131–1143, 1990.

[JNV$^+$21] Zhengfeng Ji, Anand Natarajan, Thomas Vidick, John Wright, and Henry Yuen. MIP$^* = RE$. *Communications of the ACM*, 64(11):131–138, 2021.

[Mag37] Wilhelm Magnus. Über beziehungen zwischen höheren kommutatoren. *Journal für die reine und angewandte Mathematik*, 177:105–115, 1937.

[Mer90] N. David Mermin. Extreme quantum entanglement in a superposition of macroscopically distinct states. *Physical Review Letters*, 65(15):1838–1840, 1990.

[MKS76] Wilhelm Magnus, Abraham Karrass, and Donald Solitar. *Combinatorial Group Theory: Presentations of Groups in Terms of Generators and Relations*. Dover Publications, New York, 1976.

[NPA08] Miguel Navascués, Stefano Pironio, and Antonio Acín. A convergent hierarchy of semidefinite programs characterizing the set of quantum correlations. *New Journal of Physics*, 10(7):073013, 2008.

[PGWP$^+$08] David Pérez-García, Michael M. Wolf, Carlos Palazuelos, Ignacio Villanueva, and Marius Junge. Unbounded violation of tripartite bell inequalities. *Communications in Mathematical Physics*, 279(2):455–486, 2008.

[Slo19] William Slofstra. The set of quantum correlations is not closed. *Forum of Mathematics, Pi*, 7:e1, 2019.

[The20] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, pages 367–381. Association for Computing Machinery, 2020.

[TZB+26] Ziao Tang, Chengkai Zhu, Ge Bai, Xin Wang, and Ranyiliu Chen. Formalization and exact certificates for a sharp local-question threshold in four-player XOR games. GitHub repository, 2026.

[WH23] Adam Bene Watts and J. William Helton. 3XOR games with perfect commuting operator strategies have perfect tensor product strategies and are decidable in polynomial time. *Communications in Mathematical Physics*, 400:731–791, 2023.

[WHK23] Adam Bene Watts, J. William Helton, and Igor Klep. Noncommutative Nullstellensätze and perfect games. *Annales Henri Poincaré*, 24:2183–2239, 2023.

[WHKN19] Adam Bene Watts, Aram W. Harrow, Gurtej Kanwar, and Anand Natarajan. Algorithms, bounds, and strategies for entangled XOR games. In *10th Innovations in Theoretical Computer Science Conference*, volume 124 of *Leibniz International Proceedings in Informatics*, pages 10:1–10:18, 2019.

[WW01] Reinhard F. Werner and Michael M. Wolf. All-multipartite bell-correlation inequalities for two dichotomic observables per site. *Physical Review A*, 64(3):032112, 2001.

[Zha24] Yuming Zhao. Robust self-testing for nonlocal games with robust game algebras, 2024.

[ZTZ+26] Chengkai Zhu, Ziao Tang, Guocheng Zhen, Yimeng Cao, Yusheng Zhao, Ranyiliu Chen, Xuanqiang Zhao, Lei Zhang, and Xin Wang. Lean-QIT: Towards a formal infrastructure for quantum information theory, 2026.

A Lean correspondence

The accompanying development [TZB+26] formalizes the principal threshold theorem and its obstruction-space formulation in Lean 4 [dMU21], using Mathlib [The20] and the Lean-QIT library [ZTZ+26]. Lean-QIT is imported as the external Lake package `QIT`; it is not vendored into the accompanying repository. Figure 1 displays the dependency graph for the two parts of Theorem 1.1. Table 2 gives the statement-by-statement correspondence.

The status *direct* means that the Lean declaration states the manuscript conclusion up to notation and packaging. *Partial* marks a proper part of a manuscript statement, *supporting* marks a conditional component or a consequence used by a directly formalized result, and *alternative* marks an application-specific combinatorial route in place of the more general lemma stated in the manuscript. This distinction is material for the exact-multiplicity circuit statement, the refutation clause of Theorem 4.1, and the general Magnus lemma: their downstream conclusions are kernel checked, but the labeled intermediate statements are not each reproduced verbatim in Lean.

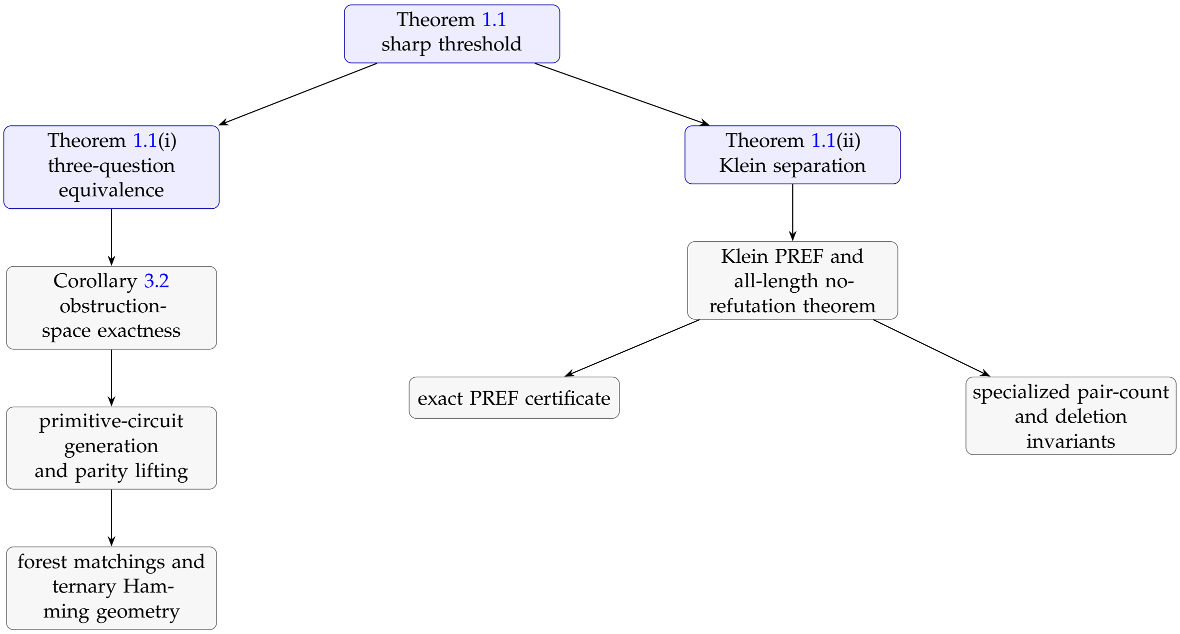

Figure 1: Core dependency graph for the Lean formalization. Arrows point from a conclusion to its principal ingredients.

Table 2: Correspondence between manuscript statements and Lean declarations.

Manuscript statement	Principal Lean declaration(s)	Status
Theorem 1.1(i)	`QIT.XORGame.commutingOperatorValue_eq_one_iff_hasPerfectMERP_of_activeQuestionBound_le_three`	Direct
Theorem 1.1(ii) and sharpness	`QIT.XORGame.v4_value_one`; `QIT.XORGame.v4_no_MERP`; `QIT.XORGame.four_is_sharp_threshold`	Direct
Proposition 2.1	`QIT.XORGame.commutingOperatorValue_eq_one_iff_of_support_eq`; `QIT.XORGame.weightedValue_lt_one_of_uniformValue_lt_one`	Partial
Theorem 3.1	`QIT.XORGame.primitiveCircuit_lifts_fourByThree`	Supporting
Corollary 3.2	`QIT.XORGame.obstructionSpaces_eq_fourByThree`	Direct
Lemma 3.3	`QIT.XORGame.balancedWord_alternatingLift_isIntegerRelation`; `QIT.XORGame.alternatingLift_mod_two`	Direct
Proposition 3.4	`QIT.XORGame.primitiveCircuits_closure_eq_kernel`; `QIT.XORGame.everyIntegerRelationHasBalancedLift_of_primitiveCircuits`	Direct
Lemma 3.5	`QIT.XORGame.fullCircuit_questionFiber_card_ne_one`	Direct
Proposition 3.6	`QIT.XORGame.fullRationalCircuit_card_eq_rank_add_one`; `QIT.XORGame.rationalIncidence_rank_le_usedQuestions`; `QIT.XORGame.fourByThree_primitiveCircuit_card_le_ten`	Direct
Lemma 3.7	`QIT.XORGame.parityPairing_occurrenceParity`; `QIT.XORGame.exactMultiplicityBalancedLift_hasBalancedLift`	Supporting
Lemma 3.8	`QIT.XORGame.InvolutionWord.reducesToEmpty_of_toFinset_card_le_two_of_alternatingSum_zero`; `QIT.XORGame.exists_bipartiteLinearForestAlternatingLayout`	Direct

Continued on next page

Table 2 (continued)

Manuscript statement	Principal Lean declaration(s)	Status
Proposition 3.10	`QIT.XORGame.selectedBlockLinearForest_exactMultiplicityBalancedLift`	Direct
Proposition 3.11	`QIT.XORGame.threeColor_colorwiseMatching_linearForest`	Direct
Proposition 3.12	`QIT.XORGame.ColorSignatureSystem.linearForestMatching_of_singleton`	Direct
Proposition 3.13	`QIT.XORGame.hasExactMultiplicityBalancedLift_of_ternaryPlayers_card_le_three`	Direct
Lemma 3.14	`QIT.XORGame.safeCenters_card_le_six_of_even_separated`; `QIT.XORGame.evenSupportDistances_separated_of_sharedBlockMissingCell_escape`	Partial
Lemma 3.15	`QIT.XORGame.safeHole_neighborhood_excess`	Direct
Proposition 3.16	`QIT.XORGame.atMostEight_exactLift_of_nonliftable_even_separated`	Supporting
Lemma 3.17	`QIT.XORGame.nineBallCover_rationalIncidence_linearIndependent`	Direct
Proposition 3.18	`QIT.XORGame.nineClause_exactLift_of_nonliftable_even_separated`	Supporting
Proposition 3.19	`QIT.XORGame.tenClause_noncovering_geometry_impossible`	Supporting
Proposition 3.20	`QIT.XORGame.tenBallCover_structural_contradiction`; `QIT.XORGame.tenBallCover_not_fullRationalCircuit`	Supporting
Theorem 4.1	`QIT.Research.FourXOR.hasCommonNoncrossingOrder_iff_isCyclic`; `QIT.Research.FourXOR.cyclic_cayley_has_true_refutation`	Partial
Lemma 4.2	`QIT.Research.FourXOR.NestedWord.equalPairs_noncrossing`; `QIT.Research.FourXOR.NestedWord.reducible`	Supporting
Lemma 4.3	`QIT.Research.FourXOR.same_side`	Direct
Lemma 4.4	`QIT.Research.FourXOR.S_succ_dj_eq_d`; `QIT.Research.FourXOR.S_pow`	Partial
Lemma 4.5	reduction interfaces in `FourXOR.AllLength` and `FourXOR.AllLengthGame`	Alternative
Lemma 4.6	`QIT.Research.FourXOR.altSum_reducible`; `QIT.Research.FourXOR.pairSum_reducible`	Alternative
Elementary-abelian all-length conclusion	`QIT.Research.FourXOR.cayley_f2r_hasPREF_and_no_refutation`	Direct

The source tree contains no `sorry`, `admit`, or manually declared axiom. The repository additionally runs `#print axioms` on the paper-facing declarations. Their dependency sets contain Lean's standard `propext`, `Classical.choice`, and `Quot.sound`. The three-question circuit argument also uses seven explicitly enumerated `native_decide` bridge axioms for finite cardinality and Hamming-space computations; the Klein witness, translation matching classification, and elementary-abelian all-length theorem use only the three standard axioms. The tracked toolchain, Lake manifest, exact certificate scripts, and full declaration map are included in the accompanying repository [TZB+26].